

THE EMERGENCE OF COMPLEX BEHAVIOR IN LARGE-SCALE ECOLOGICAL ENVIRONMENTS

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ABSTRACT

We establish how physical scale and population size govern the emergence of complex behavior in open-ended ecological environments. In our setting, agents are unsupervised—endowed with neither explicit rewards nor learning objectives—and instead evolve over time through the coupled dynamics of reproduction, mutation, and selection. Because agents continually reshape both their environment and the surrounding population, the system constitutes an ongoing, self-modifying ecology. Rather than optimizing a single high-performance policy, we characterize how behaviors emerge and diversify across large populations under natural competition and environmental pressure. Leveraging modern hardware together with a new multi-agent simulator, we scale the environment and population well beyond previously attempted sizes, sustaining populations of over 60,000 agents, each governed by its own evolved neural network policy. Across these simulations we identify a range of emergent behaviors—including long-range resource extraction, vision-based foraging, and predation—that arise directly from competitive and survival pressures. Examining how sensing modalities and environmental scale shape the emergence of these behaviors, we find that several emerge only in sufficiently large environments and populations, and that increased scale systematically strengthens their stability and consistency. Against a rich history of research in evolutionary settings, our scaling results on modern hardware chart a promising direction for studying ecology as an instrument of machine learning in an era of increasingly abundant computational resources and efficient machine frameworks. Experimental code will be released upon acceptance.

1 INTRODUCTION

Since Darwin, the evolutionary emergence of biodiversity and complex behavior has been studied across a wide range of biological disciplines, including evolutionary theory (Darwin 1859; Simpson 1944), ecology (MacArthur & Wilson 2001), and genetics (Dobzhansky 1937). Models of these emergent dynamics span theories of speciation and adaptive radiation (Coyne & Orr 2004; Schluter 2000) and the mathematical analyses of population genetics (Fisher 1930; Wright 1931).

Within these fields, decades of mathematical modeling, laboratory experiments, and field work have profoundly advanced our understanding of the mechanisms of evolution in nature. Studying emergent behavior directly in the natural world nonetheless remains fraught with obstacles: as ecosystems grow larger and more complex, they become correspondingly harder to control and to measure, and even where controlled experiments in large-scale natural settings are feasible, they risk damaging or displacing wild populations.

To address these obstacles and better understand the emergence of complex behavior under competitive and environmental pressures, we study the open-ended evolution of neural network policies in large-scale ecological simulations. In this setting, agents possess neither a specified objective nor a reward signal; instead, they collect resources to survive and reproduce according to the dynamics of the environment, and policy changes propagate solely through mutation as parents transmit their policies to their offspring. Capitalizing on the rapid growth in computational resources available to researchers, we study open-ended evolutionary dynamics in populations of these simple digital organisms at scales not previously attainable, providing new tools for investigating how environmental scale and population size drive behavioral emergence.

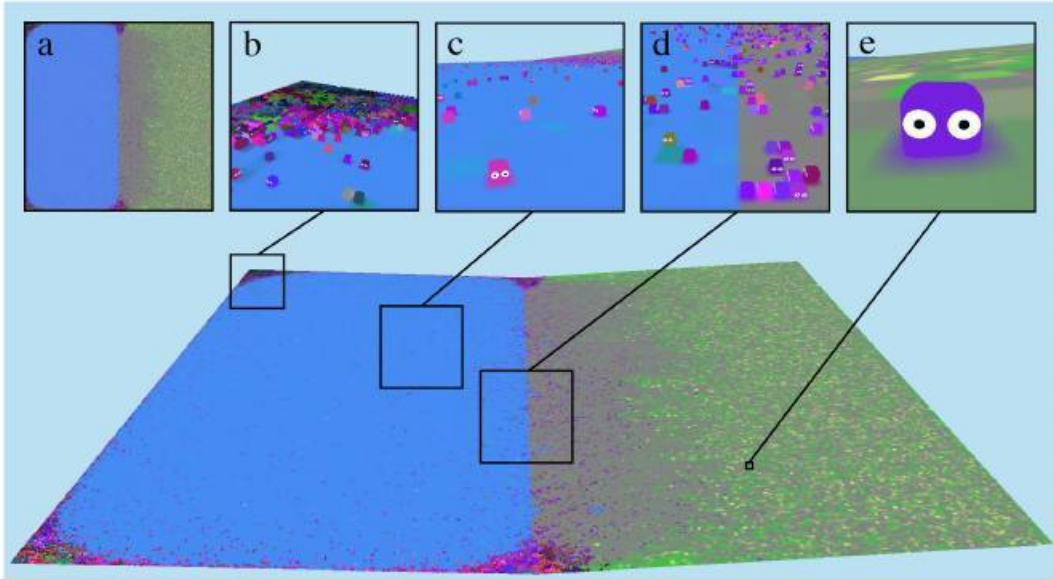


Figure 1: A 3D visualization of a 512×512 map in our environment. Inset (a) presents a top-down view of the entire map; (b) shows a group of agents clustered in a corner; (c) shows agents in the water; (d) shows agents travelling back and forth between water and land; and (e) shows a close-up of a single agent gathering resources.

To this end, we employ a new JAX-based simulation environment that supports rapid evaluation of large grid worlds. Our larger experiments comprise over 60,000 individual agents distributed across a physical area exceeding 1,000,000 grid cells. Agents in this environment must navigate to locate food and other resources in order to survive, and may reproduce by accumulating sufficient resources to instantiate a mutated copy of themselves. A 3D visualization of this environment is provided in Figure 1.

Our aim is twofold: to illuminate ecological phenomena through simulation, and to establish ecology as a mechanism for producing machine intelligence. Recent years have demonstrated that in AI, scale does not merely improve models—it fundamentally transforms their capabilities (Wei et al., 2022a). Despite this insight, embodied learning has largely remained confined to comparatively small environments with modest populations. Our work therefore asks what capabilities emerge when we scale not a single model but an entire ecosystem. Just as reasoning emerges in individual models only beyond a certain size threshold (Wei et al., 2022b), we characterize the strategies that evolve within populations only once environmental size and complexity attain sufficient richness.

Through extensive experiments, we identify multiple regimes in which sensor configurations and our large environmental scales decisively shape the behaviors that emerge from simulation, in turn producing long-term ecological consequences. For instance, we demonstrate that agents equipped with simple compass sensors reliably adapt to conduct long-range resource-gathering expeditions, a behavior that emerges far less consistently at small scales; likewise, agents endowed with vision sensors evolve markedly more effective foraging and attacking behavior than agents without them, and here too the effect strengthens with scale. Together, these experiments establish that the scale and complexity enabled by modern hardware accelerators open promising new avenues for studying how evolution in ecological environments gives rise to rich new behaviors.

2 RELATED WORK

The emergence of life has inspired computational research since the field’s inception. Conway’s Game of Life (Games, 1970) demonstrated that populations governed by simple local rules can produce intricate global structure. Early computer programs such as Tierra (Ray, 1992) and Avida (Brown) modeled life as self-replicating programs that evolve through mutation, while Reynolds

(1987), in the same spirit as Conway, derived emergent flocking and schooling behavior from local rules of motion.

Researchers have applied these principles to evolve the locomotion morphology (Sims, 1994a,b; Ha, 2018; Gupta et al., 2021; Auerbach et al., 2014; Auerbach & Bongard, 2012; Jeon et al., 2025; Auerbach & Bongard, 2014; Szerlip & Stanley, 2013) and sensing (Pratt et al., 2022; Mugan & MacIver, 2020; Tiwary et al., 2025) of virtual agents, while neuroevolution methods have evolved neural network architectures directly (Stanley & Miikkulainen, 2002, 2004; Stanley et al., 2009). Our work departs from these approaches in a fundamental respect: rather than engineering agents that better adapt to their environment under a reward or objective signal, we characterize the behaviors that emerge spontaneously from ecological pressure and quantify how environmental scale and complexity govern those behaviors.

Evolution has likewise inspired the internal dynamics of optimization algorithms. Genetic algorithms (Holland, 1992; Jong, 1975) and evolutionary strategies (Wierstra et al., 2014; Hansen & Ostermeier, 2001; Rechenberg, 1973; Schwefel, 1995; Conti et al., 2018) recruit the principles of evolution and natural selection for optimization, while related methods explicitly reward novelty among the resulting solutions (Lehman & Stanley, 2011; Cully et al., 2015). Such approaches have been used to optimize neural networks for tasks such as Atari games and Mujoco locomotion tasks (Salimans et al., 2017; Such et al., 2017; Mania et al., 2018). Although our work also employs evolutionary mechanisms to update agents, its object is the study of emergent behavior rather than the development of a novel optimization algorithm or the application of an existing one to a fixed objective.

A rich body of work likewise draws on biological inspiration to develop organisms within an environment. Early examples include Bedau (1992), Yaeger et al. (1994), Channon & Damper (1998), and Tisue et al. (2004). More recent efforts use cellular automata to grow organisms with emergent morphologies and behaviors (Mordvintsev et al., 2020; Robinson, 2021; Heinemann, 2021; Randazzo & Mordvintsev, 2023; Caussan, 2025), while Mordatch & Abbeel (2018) and Park et al. (2023) exploit simulated environments to investigate the emergence of language and communication between agents. The emergence of behavior through autocurricula (Leibo et al., 2019)—the principle that populations of agents constitute an increasingly challenging learning landscape for one another—has similarly been examined, though in environments with substantially smaller populations than ours (Leibo et al., 2018; Team et al., 2021). Complementarily, Wang et al. (2019) and Aki et al. (2024) adopt the inverse strategy, evolving environments specifically optimized for agent learning; although these works evolve the environment rather than the agent, they directly motivate our approach by demonstrating how environmental complexity elicits sophisticated behavior.

Most closely related to our work are several efforts that simulate populations of agents interacting within complex environments. Baker et al. (2019) examine a Hide-and-Seek task in which a small number of hider agents learn to exploit environmental tools to evade seeker agents, both trained via reinforcement learning. Other works similarly train populations for specific tasks: playing video games (Jaderberg et al., 2019), manipulating objects (Petrenko et al., 2021), or capturing agents (Zheng et al., 2017; Yamada et al., 2020). Although many of these works explore emergence, they do so under fixed tasks and objectives. The Neural MMO framework (Suarez et al., 2019, 2023) adopts a more open-ended posture, placing agents within a massively multi-agent game setting and optimizing only for survival rather than fixed objectives; its population, however, remains capped at a fixed number of agents (experiments examine up to 128 concurrent agents) without generational inheritance, which limits the study of population dynamics. Hamon et al. (2023) aligns more closely with our setting through variable population sizes, genetically inspired evolution, and generational timelines, yet operates in a significantly smaller and less feature-rich environment and does not directly interrogate the role of scale. Similarly, Lu et al. (2024) simulates agents evolving to gather resources, but likewise operates at a much smaller scale (a maximum of 256 agents) and emphasizes tool use through programmable robots. While each of these efforts advances the study of emergent ecological dynamics, a comprehensive account demands populations that grow and shrink naturally, evolve without predefined objectives, and interact within environments rich enough to support diverse strategies at scales where collective patterns can crystallize. Our work unifies these elements, rendering ecological phenomena that were inaccessible in prior work directly observable.

Finally, we build on recent work that emphasizes environmental speed and scale, using optimized libraries to increase the computational throughput and number of agents able to interact within game-

like environments (Koyamada et al. 2023; Matthews et al. 2024; Lechner et al. 2023). We extend this progress in computational efficiency, adopting analogous JAX-based acceleration techniques to attain large scales and fast simulation in ecological environments.

3 METHOD

3.1 ECOLOGICAL GAMES

We formalize the interactions between populations of agents and their environment as a class of multiplayer game we term an **Ecological Game (EG)**. In the manner of Markov decision processes (MDPs) in reinforcement learning and stochastic games (SGs) in multi-agent reinforcement learning (see Sutton et al. (1998) and Albrecht et al. (2024) for reference), an EG possesses underlying Markovian transition dynamics, but is expressly constructed to represent changing populations of agents in an objective-free setting. As in an SG, an EG comprises a set of agents I , a state space S , an initial state distribution ρ_0 , an action space A , and a transition function $\mathcal{T}$ mapping a state s_t together with a per-player action set $a_{t,i}$ to a distribution over subsequent states s_{t+1} . Departing from the SG formulation, an EG discards the reward function and introduces a population function $\Gamma(s_t)$ that reports which agents in I are currently alive and identifies their parents, so that birth and death events can be inferred. Just as MDPs and SGs admit partially observable counterparts—POMDPs and POSGs—an EG likewise admits a partially observable variant, which we denote a POEG. A POEG differs from an EG in that it precludes each agent from observing the entire state, instead incorporating an observation function $O(s_t, i)$ that maps a state and a player to that player’s observation. Our environment is a POEG whose observation function varies with the sensor configurations detailed in the next subsection.

3.2 ENVIRONMENT

The specific POEG used in our experiments is a large grid world populated with resources that agents must collect in order to reproduce and survive. The environment maintains a factored state space $s_t = \{s_t^m, s_t^a\} \in S = S^m \times S^a$, where s_t^m denotes the 2D data associated with the map—such as terrain and the distribution of resources—and s_t^a denotes the data associated with the agent, such as its position, health, and internal resources.

Terrain: Each map is defined by a rock layer specifying the height of every grid cell. Water is introduced into cells whose rock value falls below a preset sea level and, throughout an episode, flows downhill according to a discrete flow simulation driven by this height field. The experiments below employ six global terrain configurations, shown in Figure 2: **Ocean**, **Beach**, **Island**, **Lake**, **Isthmus**, and **Channel**. We conduct experiments at 64×64 , 128×128 , 256×256 , 512×512 , and 1024×1024 grid sizes, scaling the initial population size in proportion to the change in overall grid area as the grid grows or shrinks. Once the simulation begins, the population fluctuates freely according to the dynamics of the environment.

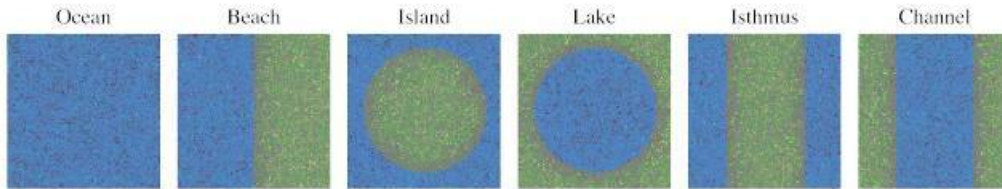


Figure 2: Overhead images of the terrains used in our experiments at the 256×256 grid size. Small dark spots denote individual agents, while green and yellow indicate concentrations of energy and biomass. Each image was captured 5000 steps into an episode—after most biomass in the water has been consumed, but before the agents have evolved to survive on land for extended periods.

Resources and Agent Health: To survive and reproduce, agents must collect three resources—**water**, **energy**, and **biomass**. Biomass and energy are initially distributed at random throughout the environment but may be consumed and redistributed by individual agents.

Every action an agent takes expends a small quantity of energy and water. Spent water is returned to the landscape, where it flows downhill to the nearest body of water, whereas spent energy is consumed and destroyed. Biomass underwrites reproduction: because every agent requires a fixed amount of biomass merely to exist, an agent cannot reproduce until it has accumulated sufficient surplus biomass to donate to its offspring. Like water, total biomass is conserved across the simulation; unlike water, biomass neither flows downhill nor moves of its own accord. Energy, by contrast, is not conserved but instead accrues slowly on free-standing biomass. We deliberately designed these coupled resource dynamics to pose a modestly harder problem than the collection of a single food source. For ablations on resource rules, see Appendix F.2

Each agent additionally possesses health points (HP) that it must maintain to avoid death. Over-spending a critical resource triggers rapid health deterioration, though an agent may expend energy and water to recover HP gradually over time. Agents may also attack one another, reducing the target’s HP, and sustain a small but monotonically increasing amount of damage as a function of age, so that an agent eventually dies of old age if no other cause intervenes. When an agent’s HP reaches zero and it dies, whatever resources it was carrying are returned to the environment in the grid cell it occupied; consequently, an agent that attacks and kills another may then consume the victim’s resources, constituting a rudimentary form of predation. For ablations on HP hyperparameters, see Appendix F.8

Sensors: Agents perceive the environment through four sensors. The **internal** sensor reports the agent’s age, health, and carried resources; the **external** sensor detects external resources within the agent’s own grid cell; the **compass**, inspired by the magnetoreceptors found in birds and other animals, encodes the agent’s global heading as a 4-way one-hot vector; and **vision** supplies a 7×7 top-down rendered patch centered on the agent, comprising three color channels together with a fourth channel encoding the local elevation gradient. For ablations on vision configurations, see Appendix F.6

Actions: Agents select from a finite set of discrete actions organized into five categories. The first is **Rest**, which performs no operation and incurs the lowest energy cost. Three **Move** actions—rotate clockwise, rotate counterclockwise, and advance one grid cell—govern locomotion; because no two agents may occupy the same cell, a move action is canceled whenever an agent attempts to enter a cell held by another agent or whenever two agents contend for the same cell. A single **Attack** action kills every agent within the 3×3 square directly ahead of the actor. For ablations on attack strength, see Appendix F.7 Three distinct **Eat** actions, one per resource type, each transfer a fixed quantity of the corresponding resource from the environment into the agent’s stomach—of fixed maximum capacity—whenever the agent occupies a cell containing that resource.

Finally, a single **Reproduce** action produces an offspring via single-parent reproduction. This action succeeds only when the agent has accumulated resources sufficient to endow its child, which is instantiated directly behind the parent. The child’s policy is a mutated copy of the parent’s weights, and the child immediately inherits a fixed fraction of the parent’s resources.

In the experiments below, we systematically disable specific sensors or the attack action in order to isolate the ecological effects of differing agent capabilities. We consider three sensing combinations: internal+external resources only (**R**); internal+external+compass (**RC**); and internal+external+compass+vision (**RCV**). The suffix **+A** indicates that attacking is enabled; for example, (**RC+A**) denotes an agent equipped with resource sensors and a compass that can also attack.

3.3 POLICIES

Agent policies are realized as small, memoryless multilayer perceptrons (MLPs); we adopt memoryless agents deliberately, to simplify the computational model and reduce the parameter count. The raw values from each sensor are flattened, normalized to the range $[-1, 1]$, and passed through separate linear layers with output dimension 64. The resulting embeddings are summed and fed through a 2-layer MLP with hidden dimension 64 and ReLU nonlinearities. Because the action space is discrete, the final layer is linear with output dimension equal to the number of admissible actions, and actions are sampled from a softmax distribution whose temperature is itself permitted to evolve alongside the network weights. Weights are stored in half-precision bfloat16 to support larger populations and faster run times. Depending on the active sensors, each agent’s network comprises roughly 10k–25k parameters. For ablations on network size and architecture, see Appendix F.5

In every run, the weights of the initial population are set by Kaiming initialization (He et al., 2015), with bias vectors initialized to zero. The policy of an agent born mid-episode is generated by copying its parent’s weights and perturbing all mutable weight values with Gaussian noise of standard deviation 3×10^{-2} , a mutation scheme consistent with prior neuroevolution research (Such et al., 2017). Although more elaborate mechanisms such as crossover are available, we deliberately restrict reproduction to this simple mutation operator so as to study the effect of environmental scale in the most parsimonious setting possible. Empirically, the low-precision floating-point representation necessitated a somewhat larger mutation rate; we ablate this choice in Appendix F.4

Each agent additionally carries a 3-channel trait governing its rendered color on the map. This color trait mutates in the same manner as the network weights and determines the agent’s appearance to other agents equipped with the vision sensor, in principle enabling agents to discriminate among distinct types of neighbors.

3.4 COMPUTATION AND REPRODUCIBILITY

All experiments are implemented in JAX (Bradbury et al., 2018) and executed on Nvidia H100 and A40 GPUs within a state-of-the-art simulation environment. Unless otherwise noted, each run proceeds for 2M environment steps or until extinction, with per-seed wall-clock times spanning under an hour for 64×64 grids to approximately 10 hours for 1024×1024 grids on a single H100 GPU. Appendix E details the simulator architecture, reports additional scaling measurements, and discusses reproducibility considerations.

4 EXPERIMENTS

The experiments below establish the effects of sensing modality, environmental scale, and population size on the emergent behavior of agents.

4.1 LONG-DISTANCE RESOURCE GATHERING

Our first experiment investigates the emergence of agents’ capacity to travel long distances inland in search of resources. We show that in environments with clear directional structure, agents equipped with a compass (**RC**) consistently develop substantially more effective long-range resource-gathering behavior than a baseline agent class (**R**) lacking a compass, and that the emergence of this behavior grows both more consistent and more effective as environmental scale increases.

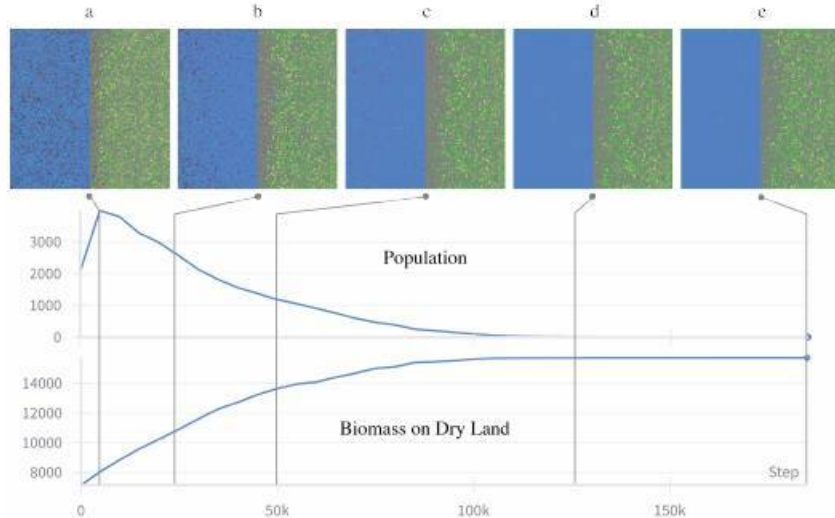


Figure 3: A representative trial with (**R**) agents in a 256×256 **Beach** environment. The top chart plots population over time; the bottom chart plots the free biomass on dry land.

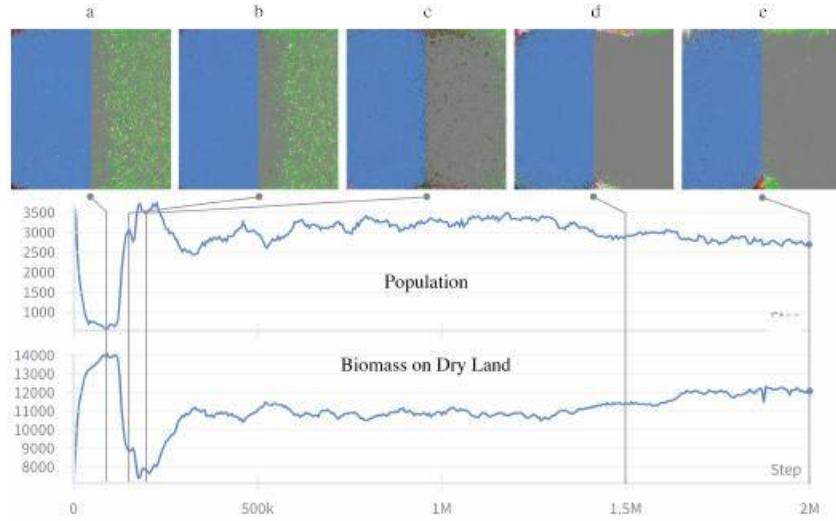


Figure 4: A representative trial with **(RC)** agents in a 256×256 **Beach** environment. The top chart plots population over time; the bottom chart plots the free biomass on dry land.

We begin with a visual example of the **(R)** agent class, shown in Figure 3. By 5000 time steps (a), all initial agents on dry land have perished for want of water, while those in the water have multiplied rapidly on the strength of the rich initial resources. At this early stage, the population has not yet evolved policies capable of efficiently seeking out resources. From this early population peak, agents in the water occasionally wander onto land (b); once there, they frequently cannot navigate back to the ocean and die, depositing their biomass on land. By 50,000 time steps (c), this gradual attrition has nearly doubled the uneaten biomass on land. The resulting transfer of resources from water to land shrinks the population, since the water no longer holds enough biomass to sustain a large population. By 125,000 steps (d), almost all biomass has migrated to land and the population is nearly extinct; the final agent dies around 185,000 steps (e).

This example lays bare the ecological consequences of the agents' inability to return to water: the deaths of individuals not only remove them from the population but also reshape the distribution of resources in ways that reverberate through the entire population. At the same time, this ecological catastrophe creates a decisive opportunity for any agent that discovers how to walk onto land, forage in the rich biomass deposits left by its dead ancestors, and safely return to water.

In the next example, shown in Figure 4, agents are additionally equipped with an internal compass **(RC)**. Initially a similar pattern unfolds (a): the population contracts as free biomass on land rapidly accumulates. In this case, however, some agents exploit their compass to drift into the relative safety of the corners. Around 150,000 steps (b), agents begin to "mine" the biomass on land, using the compass to navigate back and forth between land and water; by 200,000 steps (c) these miners have cleared the entire beach. Thereafter the population undergoes only minor fluctuations—with agents forming clumps along the edges (d)—yet remains largely stable for the full 2,000,000 time steps (e).

We quantify these observations through a full set of **Beach** experiments at grid sizes 1024×1024 , 512×512 , 256×256 , 128×128 , and 64×64 , with initial populations of 32768, 8192, 2048, 512, and 128 agents, respectively. Each configuration is run with four random seeds, and for each run we record whether mining behavior emerges—operationalized as a decline of at least 10% in the free biomass on dry land from its historical peak—together with the onset time of the mining event, determined by the local maximum in biomass on land, and the percentage drop in biomass. Table 1 summarizes the mining and extinction events, with per-seed detail in Table 3 in Appendix A. The comparison is unambiguous: mining behavior emerges consistently in **(RC)** agents but not in **(R)** agents, it emerges more consistently as environments grow larger, and it arises in every seed at and above the 256×256 scale. At smaller scales, by contrast, populations are markedly less stable and more prone to early extinction.

Table 1: Mining events in the **Beach** world for populations of agents with a compass (**RC**) versus without a compass (**R**), across five world sizes and initial populations.  denotes the number of seeds exhibiting a mining event;  denotes the number of seeds in which the population went extinct before 2M steps. Compass-equipped agents mine consistently while compass-free agents almost never do, and mining reliability rises with world size. See Table 3 in Appendix A for per-seed data.

	1024 × 1024		512 × 512		256 × 256		128 × 128		64 × 64	
	32768		8192		2048		512		128	
										
(RC)	4/4	0/4	4/4	0/4	4/4	0/4	3/4	1/4	2/4	3/4
(R)	1/4	0/4	0/4	4/4	0/4	4/4	0/4	4/4	0/4	4/4

To characterize these effects more precisely, we run additional seeds with (**R**) and (**RC**) agents at the 512×512 world size (32 seeds) and the 128×128 world size (64 seeds), and apply a Kaplan–Meier estimator to the probability of extinction and of mining as a function of simulator time. Extinction proves rarer and slower to occur at larger world sizes, whereas mining is more frequent and earlier at smaller grid sizes. Fitting a Cox proportional hazards model to the timing of mining discovery and of extinction across world sizes and compass usage corroborates these findings: larger world size substantially reduces the extinction rate and enables far more rapid mining discovery than smaller worlds, and enabling the compass produces an effect of comparable magnitude. Full details appear in Appendix B.

Table 2: Mining and extinction events in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains for populations of agents with a compass (**RC**) versus without a compass (**R**) at the 512 × 512 grid size.  denotes the number of seeds exhibiting a mining event;  denotes the number of seeds in which the population went extinct before 2M steps. Across every terrain, the compass yields more frequent mining and fewer extinctions. See Table 4 in Appendix A for per-seed data.

	Island		Lake		Isthmus		Channel	
								
(RC)	4/4	0/4	2/4	2/4	4/4	0/4	3/4	1/4
(R)	3/4	1/4	2/4	4/4	1/4	3/4	0/4	4/4

To assess how terrain configuration shapes the emergence of mining behavior, we conduct analogous experiments in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains at the 512 × 512 grid size with (**R**) and (**RC**) agents. Mining and extinction events for these experiments are summarized in Table 2 with per-run detail in Table 4 in Appendix A. As before, equipping agents with a compass yields earlier and more consistent emergence of mining behavior alongside more stable populations and lower extinction rates. (**RC**) agents adapt more readily to the island and isthmus terrains than to the lake and channel, where they occasionally go extinct; we attribute this to the fact that in the island and isthmus a simple policy of walking due east or due west always returns an agent to water, whereas the channel and lake afford no comparably simple navigational rule. The (**R**) agent fails to adapt reliably to any of these terrains, though it fares least poorly on the island. Figure 12 in Appendix D provides a visualization of mining in the island terrain.

4.2 VISUAL FORAGING AND PREDATION

We further conduct experiments in the **Ocean** environment to determine whether agents can adapt their vision sensors to locate free resources and prey on other agents effectively. We compare (**RCV+A**) agents—endowed with resource sensing, a compass, and a 7 × 7 visual sensor—against (**RC+A**) agents equipped with only resource and compass sensors.

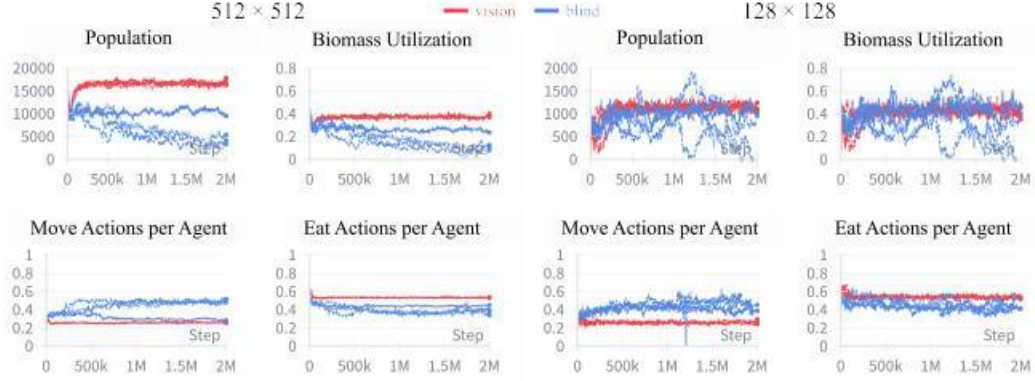


Figure 5: Population, biomass utilization, move actions per agent, and eat actions per agent in 512×512 and 128×128 **Ocean** worlds. Biomass utilization, move actions per agent, and eat actions per agent are averaged across agents at each time step. Each plot shows four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). With vision, agent populations are larger and more stable and achieve greater biomass utilization, a consequence of reduced movement together with more frequent and more effective eating. At larger scales these dynamics are more stable, with less variance across runs, indicating that in larger environments the vision sensor confers an increasing advantage over blindness in adapting foraging behavior. See Figure 9 in Appendix C for a comparison of grid sizes 1024×1024 and 256×256 .

Figure 5 plots population, biomass utilization, movement actions, and eat actions for agents at the 512×512 and 128×128 grid sizes; plots for the remaining grid sizes appear in Figure 9 in Appendix C and summary statistics in Table 5 in Appendix A. Biomass utilization is defined as the percentage of the environment’s total biomass currently held within agents. In episodes where agents possess vision sensors (red lines), they spend less time moving and more time eating, sustain larger populations, and achieve greater biomass utilization—clear evidence that the agents have adapted their visual sensing to forage more efficiently. Across environmental scales, the variance between runs decreases substantially in larger environments, and, strikingly, the vision-equipped runs exhibit markedly more stable dynamics, with less variation in these metrics over time.

Figure 6 reports the averaged attacks per agent and homicides per attack for these runs. Vision-equipped agents attack relatively infrequently yet sustain a successful kill rate of approximately 75%. In large grids, agents without vision occasionally attain high kill rates as well, but with far more unstable dynamics, and confounding factors complicate the attribution of this observation to a single cause. Visual inspection (see Figure 13 in Appendix D) reveals that these elevated kill rates often coincide with clusters of agents accumulating in one location—clusters that almost never form in runs with visual agents—which may further indicate that agents with vision adapt to avoid one another. In smaller environments the difference between visual and blind agents is less pronounced, yet populations of visual agents remain far more stable throughout.

4.3 ABLATIONS

In Appendix F we ablate specific hyperparameters—initial population size (F.1), resource rules (F.2), mutation rate (F.4), agent network configuration (F.5), vision range (F.6), attack strength (F.7), and agent HP (F.8)—to quantify the sensitivity of the emergent behaviors to simulator configuration. In Appendix F.3 we further establish that the dynamics observed in larger environments cannot be attributed merely to self-averaging.

5 CONCLUSION

Our study establishes open-ended ecological settings as a powerful paradigm for investigating the emergence of intelligence. Through experiments in worlds of varying scale and complexity, we demonstrate that both environmental scale and sensory richness contribute decisively to the devel-

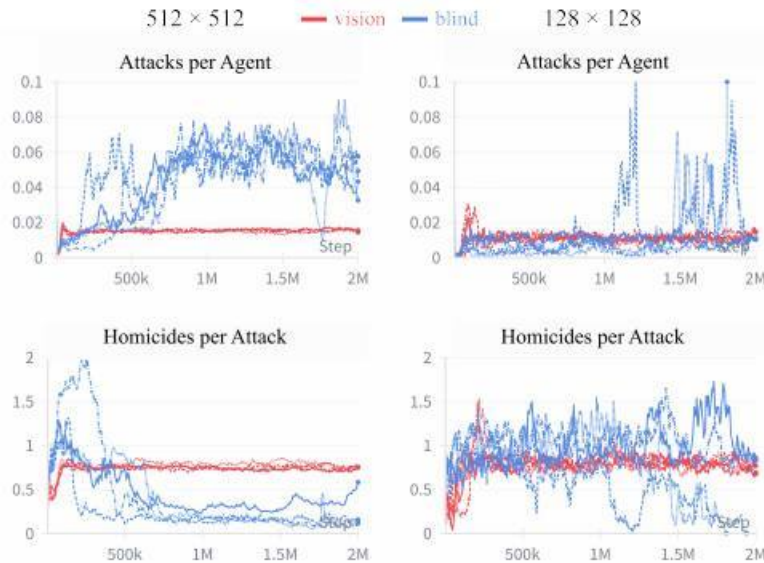


Figure 6: Attacks per agent and homicides per attack, averaged across agents, in 512×512 and 128×128 **Ocean** worlds. Each plot shows four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). With vision, agents attack less frequently but with substantially higher precision, evidencing the emergence of specialized predation behavior. This dynamic grows more pronounced at larger scales. See Figure 10 in Appendix C for a comparison of grid sizes 1024×1024 and 256×256 .

opment of sophisticated sensing and decision-making policies. Taken together, these results indicate that ecological and evolutionary pressures, when coupled with sufficiently rich environments, constitute a natural pathway toward increasingly intelligent artificial agents. Our experiments further demonstrate that rich behaviors can be evolved in large populations within a modest hardware budget and without recourse to explicit objectives. We anticipate that these results will catalyze further exploration of ecology as a mechanism for developing and understanding artificial intelligence.

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A ADDITIONAL DATA

Table 3: Per-seed mining events in the **Beach** world for populations of agents with a compass (**RC**) versus without a compass (**R**) across five scales. 🛠️ indicates whether a run exhibited a mining event. 📉 reports the percent decrease in free biomass on dry land during the mining event, and ⌚ reports the approximate timestep at which free dry biomass attained the local maximum marking the onset of the mining event; for runs with multiple mining events, we report the first. 💀 reports the timestep at which the run went extinct, or n/a if it survived the full 2M steps.

	1024x1024				512x512				256x256				128x128				64x64			
	32768				8192				2048				512				128			
Compass On (RC)																				
Seed																				
0	✓	35k	11.1	n/a	✓	105k	33.	n/a	✓	145k	49.3	n/a	✓	105k	72.5	n/a	x			55
1	✓	90k	16.2	n/a	✓	50k	17.2	n/a	✓	50k	46.6	n/a	x		140	✓	70k	85.3	675	
2	✓	40k	15.	n/a	✓	30k	14.3	n/a	✓	55k	34.7	n/a	✓	50k	50.7	n/a	✓	10k	78.	n/a
3	✓	60k	15.	n/a	✓	45k	15.6	n/a	✓	90k	47.4	n/a	✓	20k	52.	n/a	x			135
Avg	4/4	34k	14.3	0/4	4/4	58k	20.	0/4	4/4	85k	44.5	0/4	3/4	58k	58.4	1/4	2/4	40k	81.7	3/4
Compass Off (R)																				
Seed																				
0	x			n/a	x			945k	x			185k	x			65k	x			45k
1	✓	275k	12.4	n/a	x			1085k	x			110k	x			55k	x			40
2	x			n/a	x			1070k	x			165k	x			55k	x			25
3	x			n/a	x			280k	x			125k	x			90k	x			40
Avg	1/4	275k	12.4	0/4	0/4			n/a	0/4			4/4	0/4			4/4	0/4			4/4

Table 4: Per-seed mining events in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains for (RC) agents versus (R) agents. Grid size is 512×512 with an initial population of 8192.

	Island				Lake				Isthmus				Channel			
Compass On (RC)																
Seed																
0	✓	40k	94.3	n/a	x			45k	✓	45k	33.	n/a	✓	50k	27.4	n/a
1	✓	30k	94.3	n/a	✓	30k	22.5	n/a	✓	55k	37.3	n/a	✓	80k	30.8	n/a
2	✓	30k	94.8	n/a	x			45k	✓	40k	28.1	n/a	✓	50k	41.	n/a
3	✓	70k	95.1	n/a	✓	35k	25.3	n/a	✓	30k	41.5	n/a	x			80k
Avg	4/4	43k	94.6	0/4	2/4	33k	23.9	2/4	4/4	43k	35.	0/4	3/4	60k	33.1	1/4
Compass Off (R)																
Seed																
0	x			185k	x			125k	x			145k	x			100k
1	✓	75k	30.9	n/a	x			130k	x			260k	x			105k
2	✓	85k	47.	n/a	✓	75k	23.4	1400k	✓	95k	18.8	n/a	x			105k
3	✓	115k	39.5	n/a	✓	70k	24.3	1250k	x			140k	x			100k
Avg	3/4	92k	39.1	1/4	2/4	73k	23.9	4/4	1/4	95k	18.8	3/4	0/4			4/4

Table 5: Population and foraging data for Vision (RCV+A) and Blind (RC+A) agents in the **Ocean** world across scales. denotes population size; denotes biomass utilization, the percentage of the system's total biomass currently held within agents' stomachs; denotes move actions per agent, the fraction of time an agent selected a move action, averaged across agents; and denotes the corresponding proportion of eat actions. Reported values of , , , and are averages of the metric over all steps within a seed.

	1024x1024				512x512				256x256				128x128			
	32768				8192				2048				512			
Vision (RCV+A)																
Seed																
0	58354	.396	.402	.426	16413	.369	.257	.532	4195	.382	.255	.536	1126	.426	.261	.541
1	65107	.365	.257	.529	16200	.364	.256	.533	4142	.377	.258	.536	1103	.415	.262	.537
2	64735	.363	.257	.530	16560	.372	.257	.533	4269	.388	.260	.536	1041	.392	.254	.542
3	64595	.362	.256	.532	16206	.365	.255	.531	4249	.388	.259	.533	1100	.414	.257	.538
Avg	63198	.371	.293	.504	16345	.368	.256	.532	4213	.384	.258	.535	1092	.412	.259	.539
Blind (RC+A)																
Seed																
0	22734	.143	.446	.372	6549	.173	.442	.400	2036	.209	.433	.461	545	.219	.465	.427
1	23670	.151	.426	.388	6939	.180	.438	.396	3107	.321	.390	.416	1082	.444	.416	.460
2	24745	.159	.358	.426	5448	.136	.474	.374	1587	.163	.440	.433	979	.421	.404	.436
3	20004	.130	.469	.359	10354	.265	.311	.439	3164	.324	.321	.450	788	.317	.431	.477
Avg	22788	.146	.425	.386	7323	.189	.416	.402	2473	.254	.396	.440	848	.351	.429	.450

Table 6: Predation data for Vision (**RCV+A**) and Blind (**RC+A**) agents in the **Ocean** world across scales.  denotes attacks per agent—the fraction of time an agent selected the attack action, averaged across all agents—while  denotes homicides per attack, the number of agents killed per attack action. Reported values of  and  are averages over all steps within a seed. The value of 0.562 for  in seed 3 of the 128×128 (**RC+A**) run is an outlier: this seed went extinct before 2M steps, and the dwindling population inflates the attacks-per-agent metric.

	1024x1024		512x512		256x256		128x128	
	32768		8192		2048		512	
Vision (RCV+A)								
Seed								
0	.011	.776	.015	.731	.013	.784	.010	.788
1	.016	.733	.015	.741	.014	.780	.011	.793
2	.016	.707	.015	.737	.014	.750	.012	.771
3	.016	.712	.015	.780	.013	.759	.011	.765
Avg	.015	.732	.015	.747	.014	.758	.011	.779
Blind (RC+A)								
Seed								
0	.070	.218	.042	.437	.005	1.074	.016	.558
1	.054	.300	.044	.383	.016	.871	.006	1.032
2	.076	.236	.048	.238	.031	.383	.010	.979
3	.059	.203	.044	.456	.033	.446	.562	.841
Avg	.065	.239	.045	.378	.021	.693	.148	.853

B DEMOGRAPHY

We analyze rates of extinction and mining emergence as a function of world size and agents’ access to global direction through the compass sensor. Extinction is defined as the population reaching zero individuals, and mining behavior is defined exactly as in the earlier experiments (a 10% drop in free biomass on land).

Runs are stratified jointly by world size and compass access. We simulated 64 runs each of 128×128 without a compass and 128×128 with a compass, and 32 runs each of 512×512 without a compass and 512×512 with a compass. For each of the four groups, we fit a non-parametric Kaplan–Meier estimator to the time-to-extinction distribution, treating runs that did not go extinct within 2M steps as right-censored. As Figures 7 and 8 show, worlds survive longer and produce mining sooner when they are large or when agents are endowed with compass information.

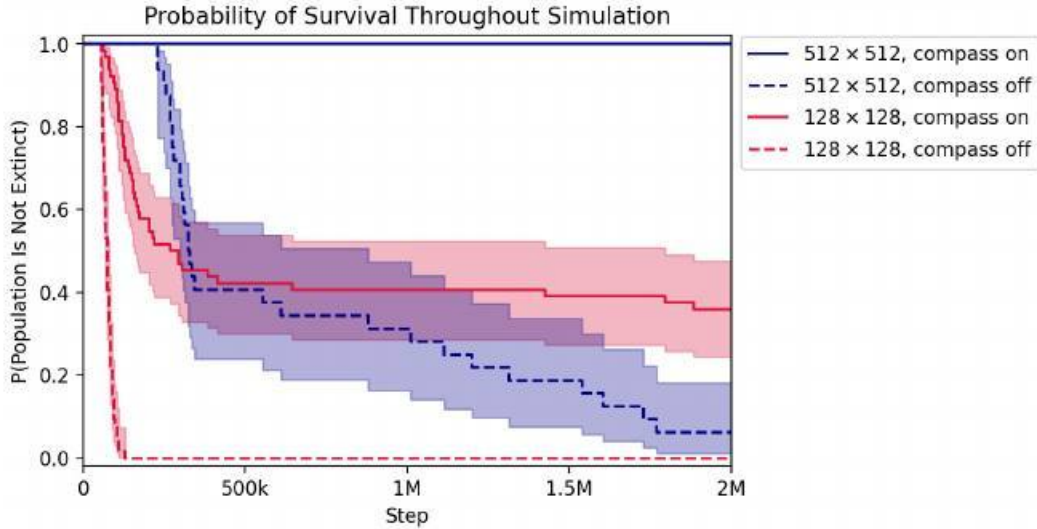


Figure 7: Probability that the population has not gone extinct, as a function of world size, compass access, and timestep. We fit a separate non-parametric Kaplan–Meier estimator for each group of runs (e.g., 512×512 , compass on); shaded regions denote the 95% confidence interval for each curve. Each 512×512 group is fit on 32 runs and each 128×128 group on 64 runs.

We compare emergence and extinction timing with a Cox proportional hazards model—a semi-parametric survival model relating covariates to event hazard rates (Table 7)—treating runs that experienced neither an emergence nor an extinction event within 2M time steps as censored. For extinction, both larger world size and compass use substantially reduce the hazard: the extinction rate in 128×128 worlds is roughly 25 times that of 512×512 worlds ($HR=0.04$), and enabling the compass produces an effect of comparable magnitude ($HR=0.03$). For mining emergence, larger worlds generate mining behavior far more rapidly ($HR=6.07$), and compass use accelerates emergence dramatically ($HR=34.59$). Both models discriminate well between groups (concordance 0.80 for extinction, 0.85 for mining). We conclude that larger worlds sustain larger populations, reducing the probability of reaching the absorbing zero boundary while simultaneously exploring a larger phenotype space—owing to the greater volume of mutation—than smaller worlds.

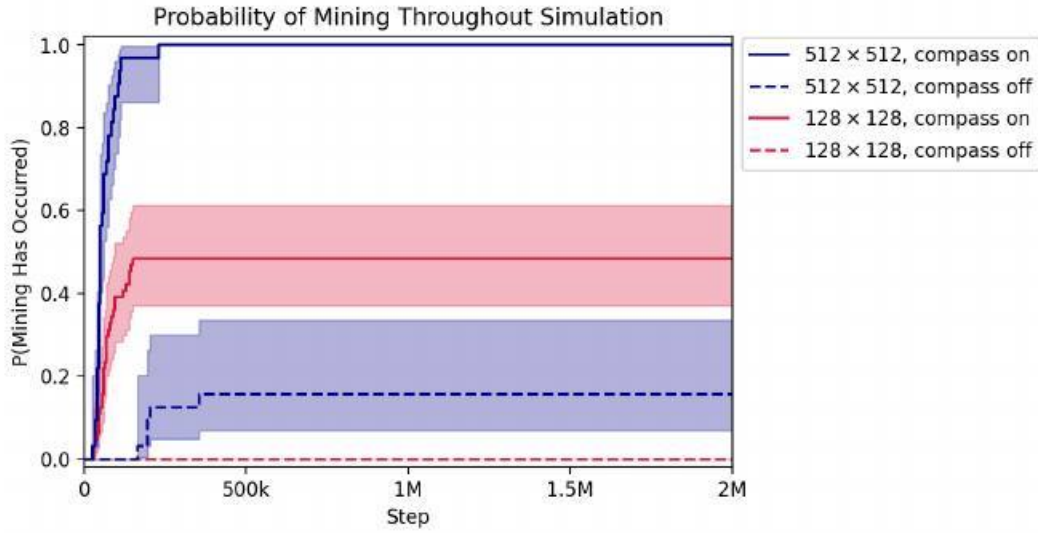


Figure 8: Probability that mining has evolved within the population, as a function of world size, compass access, and timestep. We fit a separate non-parametric Kaplan–Meier estimator for each group of runs (e.g., 512×512 , compass on); shaded regions denote the 95% confidence interval for each curve. Each 512×512 group is fit on 32 runs and each 128×128 group on 64 runs.

Table 7: Cox proportional hazards models for extinction and emergence events, relative to a 128×128 , compass-off baseline. Hazard ratios (HR) below 1 indicate reduced hazard (a longer time to event); for example, the extinction $HR = 0.04$ for 512×512 means extinction occurs at 0.04 times the rate observed in 128×128 worlds. The analysis comprises 192 runs in total: 32 seeds with compass on and 32 with compass off in the 512×512 world, and 64 seeds with compass on and 64 with compass off in the 128×128 world.

Covariate	Extinction			Mining		
	HR	95% CI	p-value	HR	95% CI	p-value
Size 512×512	0.04	0.02–0.07	< 0.005	6.07	3.61–10.20	< 0.005
Compass On	0.03	0.01–0.05	< 0.005	34.59	13.51–88.57	< 0.005
Concordance	0.80			0.85		
Events / Total	135 / 192			68 / 192		

C ADDITIONAL FORAGING AND PREDATION RESULTS

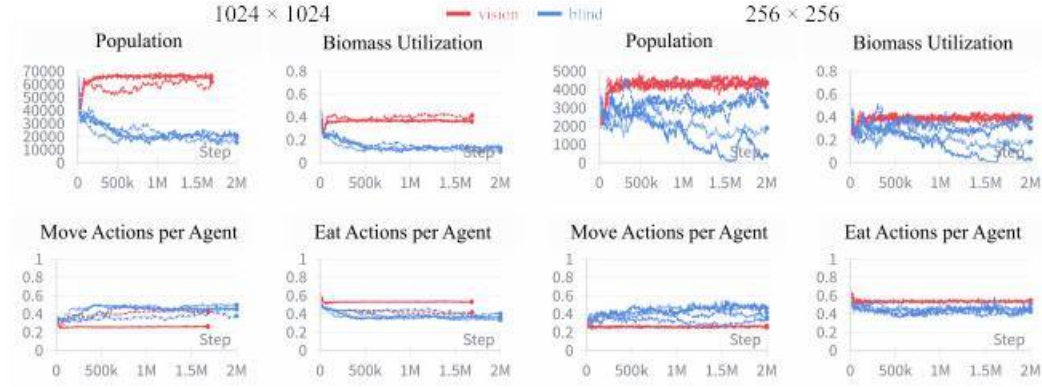


Figure 9: Foraging behaviors in the **Ocean** world across scales. Enabling vision (red) yields more stable populations and greater biomass utilization, driven by more frequent and more effective eating together with reduced movement. At larger scales these dynamics are more stable, with less variance across runs. Owing to limits on GPU hours, the vision seeds at the 1024×1024 scale ran for 1.7M rather than 2M steps.

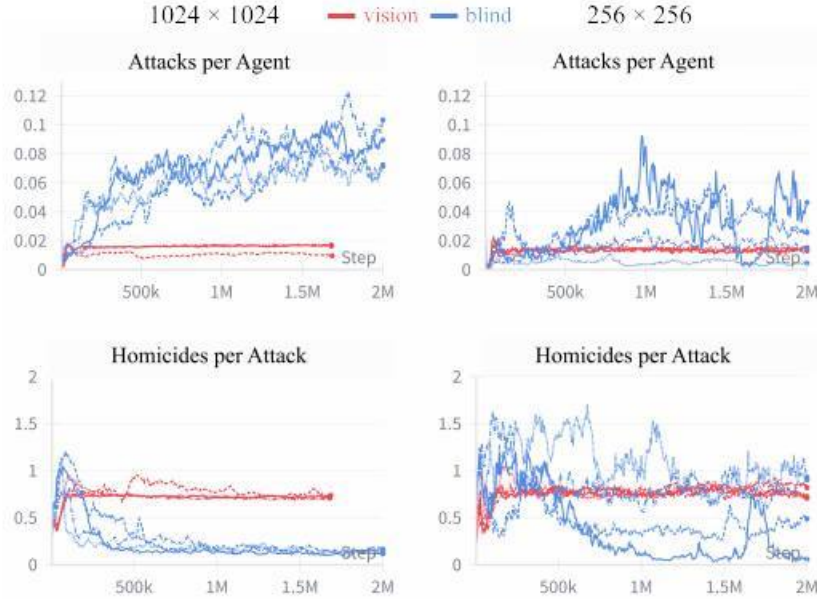


Figure 10: Predation behaviors in the **Ocean** world across scales. With vision (red), agents attack less frequently but with substantially higher precision, evidencing the emergence of specialized predation behavior alongside foraging capabilities (Figure 9). This dynamic is again more pronounced at larger scales, with greater variance in the behavior of blind agents (blue) at smaller world sizes.

D ADDITIONAL EXPERIMENT VISUALIZATIONS

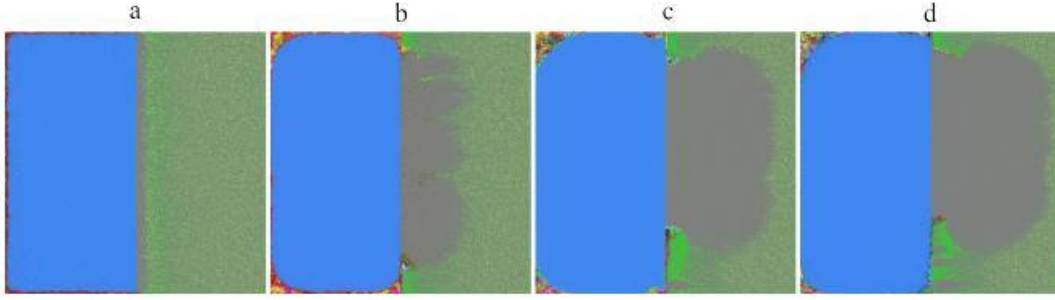


Figure 11: Larger-scale environments induce more varied and complex ecologies that give rise to distinctive visual patterns. In this 1024×1024 **Beach** world, **(RC)** agents initially died at the shore (a), transferring biomass to dry land; they subsequently developed mining behavior, clearing biomass from the beach (b). As biomass re-accumulated at the ends of the beach (c), subpopulations re-evolved mining behavior, traveling inland along these secondary biomass beaches to extract the resources (d).

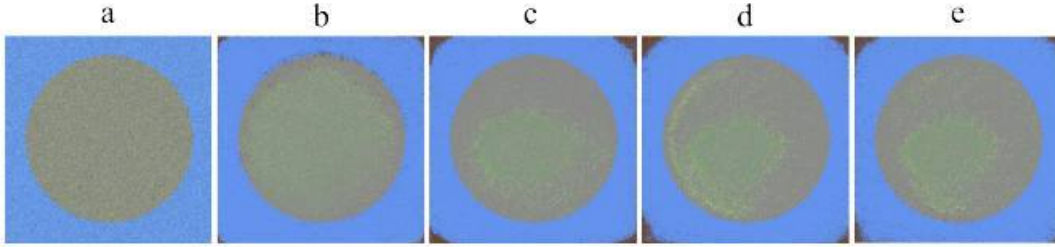


Figure 12: Mining behavior emerges across varied terrains. In this 512×512 **Island** world with **(RC)** agents, distinct regions of the shore are mined over the course of the run (b, c). Biomass is transferred back onto parts of the island once mining becomes difficult and agents die before reaching water (d); agents then adapt to re-mine this accumulated wall of biomass, clearing it from the shore (e).

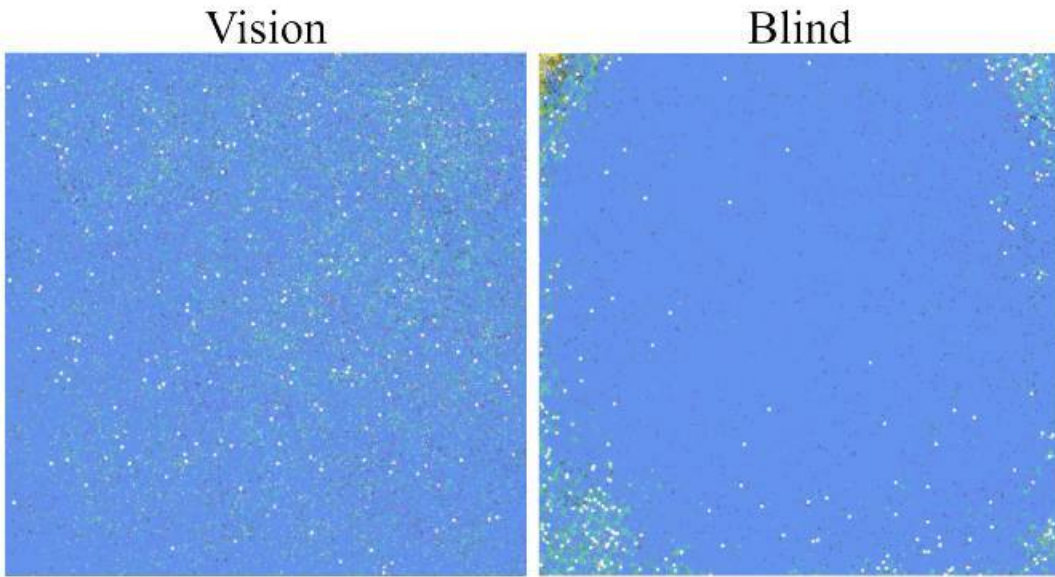


Figure 13: Representative runs in a 512×512 **Ocean** world with attacking enabled: vision **(RCV+A)** agents on the left, blind **(RC+A)** agents on the right. White squares mark locations where an agent is attacking. With vision, agents adapt to locate free resources throughout the ocean world and prey on other agents far more effectively (see Figures 5 and 6). Blind agents explore the ocean for resources to a much lesser extent, instead adopting policies that cluster in the corners of the world.

E DETAILS ON COMPUTATION AND REPRODUCIBILITY

We provide details on simulator design, runtime, and experiment reproducibility.

Simulator design. All experiments are conducted with a JAX-based, GPU-accelerated simulator purpose-built for large-scale multi-agent ecosystems with open-ended populations. The simulator computes every component of the environment step on the GPU—terrain dynamics, resource flows, agent updates, and measurement alike—so that policy evaluation and simulation share a single JAX compilation pipeline. It exposes a deliberately minimal population-level interface: given the current map state, agent states, and heritable traits, a population algorithm supplies actions and updated traits while the simulator handles births, deaths, and environmental transitions. This separation of concerns allows a single environment implementation to be reused across different evolutionary algorithms, reinforcement learning methods, and other population update rules. Because all operations are implemented as pure JAX functions combined with vectorization and just-in-time compilation, they fuse into a small number of GPU kernels, delivering the throughput required for our scaling experiments; runtime measurements for a single H100 GPU are reported in Table 8.

A central feature of the simulator is its built-in tooling. A measurement API computes and records aggregate statistics at each step—biomass utilization, movement and eating rates, attack and homicide rates, and extensive environment-level parameters—which feed directly into our figures and tables, while an interactive 3D viewer renders terrain, water, resources, and agents from generated reports and supports inspection of individual agents (Figure 1). These tools proved indispensable for discovering and validating qualitative phenomena, such as long-range resource mining and predation patterns, that are not apparent from scalar metrics alone.

From a high-performance-computing standpoint, our simulator fills a distinct gap in the existing landscape. General-purpose agent-based frameworks such as MASON (Luke et al., 2005), FLAME-GPU (Richmond et al., 2024), and Griddly (Bamford et al., 2020) offer high configurability but demand substantial environment engineering and do not directly target the open-ended, evolving populations at the scales we study; GPU-native multi-agent RL environments, conversely, emphasize fixed agent counts and task-specific reward structures. By uniting GPU-native execution, explicit support for births and deaths with mutable traits, and configurable resource and terrain dynamics, our simulator sustains open-ended populations of over 10^5 agents while remaining interoperable with diverse population algorithms. This combination constitutes a state-of-the-art platform for scaling ecological experiments such as those reported here, and a robust foundation for future work on emergent behavior in large populations. Our simulator will be released upon acceptance.

Runtime data. Table 8 reports runtime across grid sizes and maximum agent populations, with each run conducted on a single H100 GPU for a maximum of 2M steps. Although the table reports runtimes for (RC) and (RCV + A) agents with 2 MLP layers and 64 channels per layer ($\sim 22k$ parameters total), runtime varies with agent network size, the set of active sensors, and the level of logging.

Our largest runs, at the 1024×1024 grid size, require ~ 10 hours in the **Beach** terrain and ~ 14 hours in the **Ocean** terrain on one H100, with memory usage of roughly 60GB. **Ocean** runs are somewhat slower because both the agent attack action and vision sensing are enabled.

Table 8: Runtime in the **Beach** and **Ocean** worlds across grid sizes and maximum agent populations. Each (RC) agent carries a 22,465-parameter neural network and each (RCV + A) agent a 22,529-parameter network. We report simulation steps per second (hz) and the total extrapolated wall-clock runtime in hours for 2M steps; effective runtime is lower for seeds whose populations go extinct before 2M steps.

Environment	1024×1024		512×512		256×256		128×128		64×64	
	hz	runtime	hz	runtime	hz	runtime	hz	runtime	hz	runtime
Beach (RC)	55	10.1 hrs	190	2.9 hrs	315	1.8 hrs	535	1.0 hrs	750	0.7 hrs
Ocean (RCV + A)	40	13.9 hrs	142	3.9 hrs	235	2.4 hrs	415	1.3 hrs	750	0.7 hrs

Reproducibility. Because our environments are large and randomized, the precise outcome of any given experiment can vary substantially from run to run, and we therefore report four distinct random seeds for each experiment. Owing to nondeterminism in certain low-level GPU operations, repeating an experiment with an identical random seed does not reliably reproduce the same trajectory: modern GPUs may execute parallel floating-point operations such as summations in different orders across runs and fuse operations in ways that alter rounding behavior, so that the minute discrepancies introduced by finite-precision arithmetic accumulate over many steps into appreciable drift. The use of 16-bit weights coarsens these rounding effects but is not their principal source. Despite this run-to-run variation, clear high-level patterns emerge robustly once the results of multiple seeds are aggregated.

F ABLATIONS

We present a series of ablations that probe the effects of specific hyperparameters and establish the robustness of our results across environment and agent configurations.

F.1 INITIAL POPULATION SIZE SWEEP FOR FIXED MAP SIZE

We evaluate how initial population size affects the consistency of mining behavior and population stability in a 256×256 **Beach** world. Reported results are for **(RC)** agent populations; the compass-free baseline **(R)** exhibited a 0% mining rate and a 100% extinction rate across all initial population sizes except the largest tested, 8192, which yielded 25% mining and 75% extinction. Note that mining and extinction frequencies need not sum to 100%: a population may develop mining and subsequently go extinct, or may survive the full 2M steps without ever developing mining behavior.

As Figure 14 shows, initial population size has no significant effect on either mining consistency or extinction rate: populations seeded with only 128 agents evolve mining as frequently as those seeded with 8192 agents, and extinction rates remain largely invariant across initial population size.

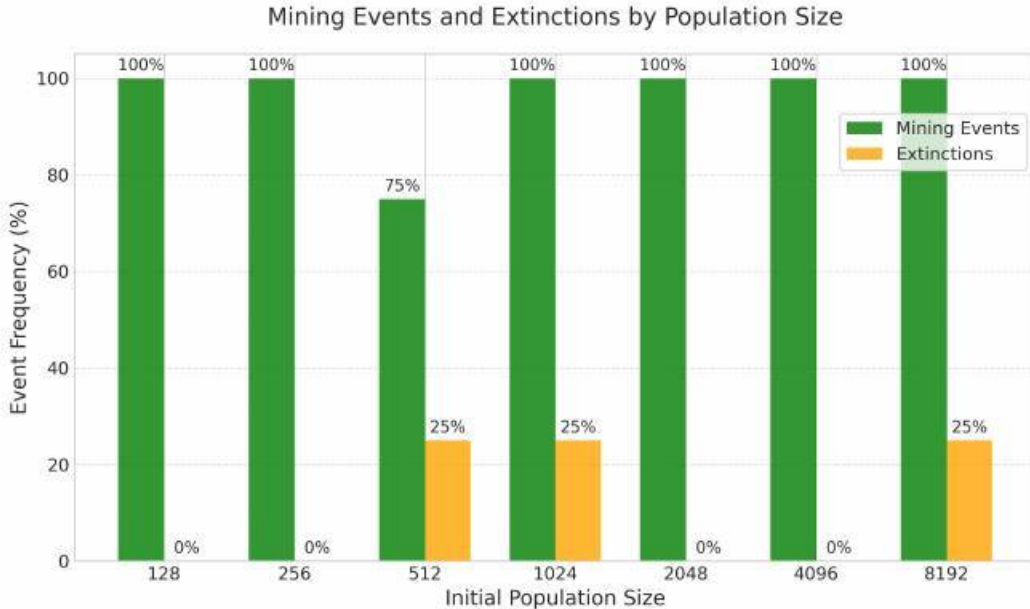


Figure 14: Frequency of mining and extinction events in a 256×256 **Beach** world across initial sizes of **(RC)** agent populations. Frequencies are computed over four random seeds per initial population size; e.g., a 25% extinction-event frequency means the population reached zero before 2M simulation steps in 1 of 4 seeds.

Figure 15 clarifies why this effect is so slight. Irrespective of its initial size, each run converges rapidly to a similar population within the first 50,000 steps of simulation. This indicates that the

environment implicitly defines a carrying capacity determined by the fixed resources in the system: populations initialized below the carrying capacity grow toward it, while populations that are too large are constrained down to it. Consequently, the effective population size is comparable across runs once the initial size has been corrected, leaving mining and extinction frequencies unaffected. Larger initial populations do, however, contribute somewhat to the total resources in the system, since agents carry biomass that is deposited upon death; as a result, larger initial populations reach higher maximum populations on average over the course of simulation (Table 9), though the effect is too small to alter mining or extinction frequency in this setting. Finally, although population size converges early, it does not remain constant thereafter: later in the simulation, populations across runs can diverge substantially, as the ecological consequences of agents evolving different abilities at different times reshape the later resource distributions.

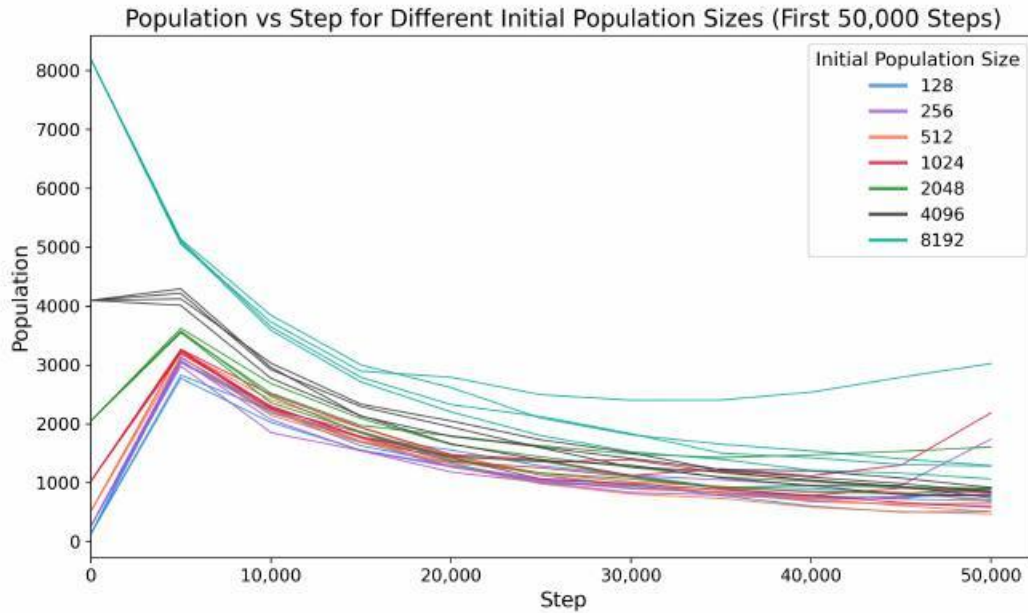


Figure 15: Population size during the first 50,000 time steps in the 256×256 **Beach** with (RC) agents, with four seeds per initial population size. The environment sustains an implicit carrying capacity set by the resources available in the system, and every initial population size approaches this value early in the run. Larger initial populations add somewhat to the total resources in the system through the biomass deposited on agent death, modestly raising the carrying capacity.

Table 9: Maximum population size reached after step 0 for different starting population sizes, with means and standard deviations across four random seeds. Larger initial populations reach slightly higher maximum populations, with greater cross-seed variance, than smaller ones; nonetheless, the maximum population reached is largely robust to the initial size, in that an initial population of 128—only 1.6% of an initial size of 8192—attains 78.9% of the latter’s maximum population (3178 vs. 4026).

Initial Population Size	Max Population Size After Step 0
128	3178 ± 217
256	3056 ± 61
512	3451 ± 437
1024	3613 ± 381
2048	3786 ± 431
4096	3689 ± 437
8192	4026 ± 642

F.2 VARYING RESOURCE RULES

We evaluate the effect of two resource hyperparameters—the biomass photosynthesis rate and the initial biomass site density—on mining and extinction frequency, in a setting otherwise identical to that of Appendix F.1. Here the initial population size is fixed at 2048 and agents again possess a compass (RC).

The biomass photosynthesis rate governs the rate at which landscape biomass creates energy via photosynthesis, replenishing resources for agents over time, while the initial biomass site density determines the fraction of grid cells containing biomass at the start of simulation. All experiments in the main body (Section 4) use a biomass photosynthesis rate of 0.01 and an initial biomass site density of 1/4.

Whereas varying the initial population size left mining and extinction consistency unchanged (Figure 14), varying the resource rules exerts a pronounced effect, as Figures 16 and 17 demonstrate.

The mechanism is intuitive: larger biomass photosynthesis rates or initial biomass site densities directly raise the effective carrying capacity of the environment by increasing the resources available in the system, so that larger populations are sustained throughout the run, which in turn increases the likelihood of mutations favoring mining behavior while diminishing the chance of extinction. Our data bear this out: averaging over the surviving seeds (omitting those that went extinct), runs with initial biomass site density 1/16 reached a maximum population of 1608 after step 0, rising to mean maxima of 2466 at density 1/8, 3786 at density 1/4, and 5353 at density 1/2.

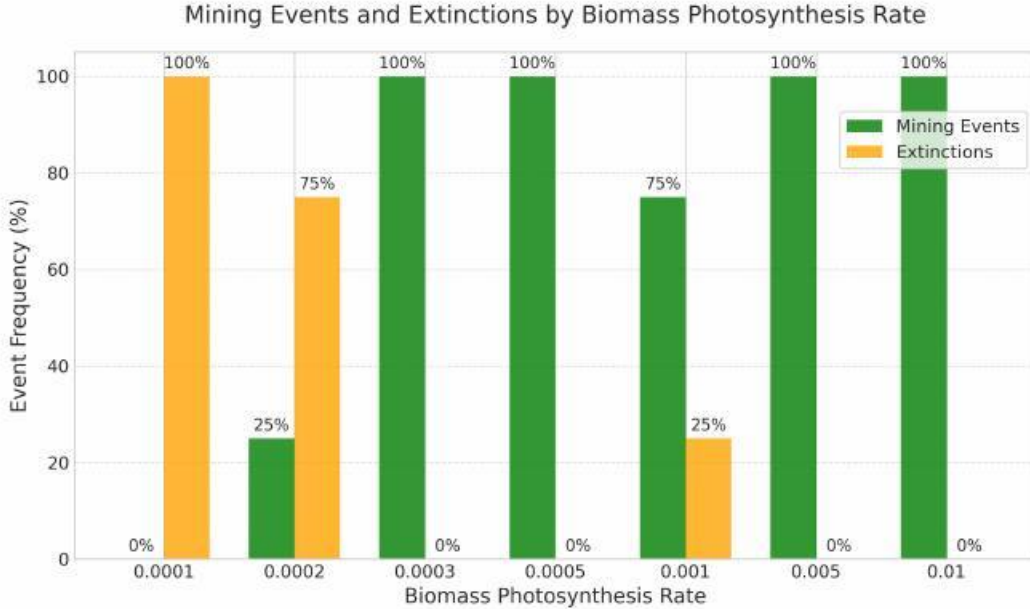


Figure 16: Frequency of mining and extinction events in a 256×256 **Beach** world across settings of the biomass photosynthesis rate. Frequencies are computed over four random seeds per setting. Mining emerges consistently at biomass photosynthesis rates of 0.0003 and above.

F.3 RULING OUT SELF-AVERAGING IN LARGER POPULATIONS

One might worry that the increased stability of mining behavior in large environments (Table 1) reflects mere self-averaging rather than qualitatively different dynamics. By self-averaging we mean the statistical-physics intuition that, as system size grows, fluctuations in macroscopic quantities shrink roughly as $1/\sqrt{N}$, where N is the system size, so that a single large system behaves like the average of many smaller systems of the same total size. On this view, a 512×512 **Beach** world would merely act as many loosely coupled 128×128 worlds run in parallel, and our scaling results could in principle be reproduced by running enough small worlds and averaging over random seeds.

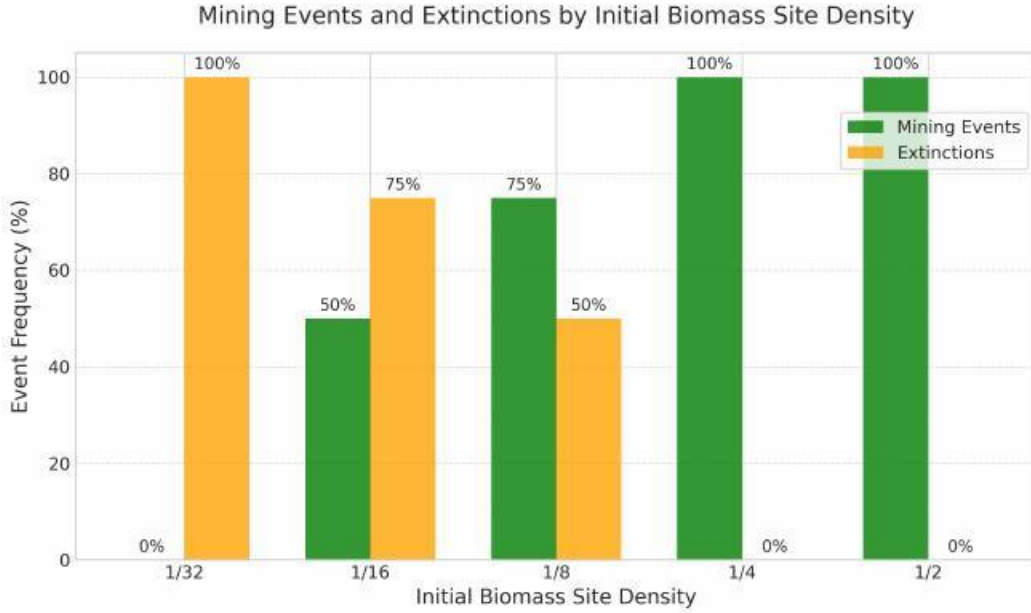


Figure 17: Frequency of mining and extinction events in a 256×256 **Beach** world across settings of the initial biomass site density. Frequencies are computed over four random seeds per setting. Higher initial biomass site densities yield more consistent discovery of mining and a lower chance of extinction.

To test this hypothesis, we hold the total “sample size” approximately fixed while varying how it is partitioned across environments. In the **Beach** terrain with compass-equipped agents (**RC**), we consider three conditions:

- 4 runs of a 512×512 world with 8192 initial agents each,
- 16 runs of a 256×256 world with 2048 initial agents each,
- 64 runs of a 128×128 world with 512 initial agents each.

In each case, the total number of grid cells across runs, the total starting resources and the total initial number of initial agents are identical:

$$4 \times 512^2 = 16 \times 256^2 = 64 \times 128^2, \quad 4 \times 8192 = 16 \times 2048 = 64 \times 512.$$

Each seed is run for 2M steps with the same hyperparameters as in Section 4.1 and, as a control, we repeat the protocol for agents without a compass (**R**). Table 10 summarizes the frequency of long-distance mining and extinction in each configuration.

Were large environments merely self-averages of many smaller ones, mining and extinction rates would be roughly invariant to how grid cells and agents are distributed across runs, and 4 large worlds would exhibit essentially the same probability of discovering mining as 64 small ones. Instead, we observe a strong dependence on world size. For (**RC**) agents, mining emerges in every run with no extinctions at 512×512 , remains common but not guaranteed at 256×256 , and becomes unreliable and fragile at 128×128 , where many populations go extinct before reaching 2M steps. The (**R**) agents, by contrast, never develop mining and go extinct at every scale, confirming that the compass remains a key enabler of long-range behavior.

These results are incompatible with a self-averaging explanation. On the contrary, larger environments demonstrably enable and stabilize mining behavior in ways that cannot be reproduced by averaging over a greater number of small simulations.

Even with the total number of cells, agents, and simulation steps matched across conditions, concentrating resources into a few large worlds yields a regime in which mining is consistent and extinctions

Table 10: Mining and extinction events under a trade-off between world size and the number of independent runs. We compare **Beach** worlds at grid sizes 512×512 , 256×256 , and 128×128 with initial populations scaled proportionally (8192, 2048, and 512 agents, respectively). For each size, the same total number of grid cells and initial agents is allocated across all runs by running 4 seeds at 512×512 , 16 seeds at 256×256 , and 64 seeds at 128×128 . We report the fraction of runs in which mining emerged (🔍) and the fraction in which the population went extinct before 2M steps (💀), for agents with a compass (**RC**) and without (**R**).

	512×512 8192		256×256 2048		128×128 512	
	🔍	💀	🔍	💀	🔍	💀
(RC)	4/4	0/4	15/16	1/16	31/64	41/64
(R)	0/4	4/4	0/16	16/16	0/64	64/64

are rare, whereas dispersing the same resources across many small worlds yields populations that frequently fail to discover mining and are far more prone to collapse. Increasing environmental scale therefore does not merely smooth out stochastic noise; it alters the underlying ecological dynamics—for instance, by supporting larger, more spatially distributed populations and more persistent diversity among evolving policies.

F.4 MUTATION RATE SWEEP

We test the sensitivity of mining emergence to the mutation rate hyperparameter, which sets the scale of the Gaussian noise added to a child agent’s policy at birth and thereby determines how much a child differs from its parent. We again run experiments in a 256×256 **Beach** world with an initial population of 2048 (**RC**) agents for a maximum of 2M steps.

Figure 18 shows that between mutation rates of 6×10^{-3} and 7.875×10^{-2} , mining emerges in at least 3 of the 4 seeds, establishing that our Section 4.1 results at mutation rate 3×10^{-2} are robust throughout this interval. As expected, mutation rates that are too small (e.g., 3×10^{-3}) prevent agents from developing new behavior and impair survival, culminating in extinction, while mutation rates that are too large (e.g., 3×10^{-1}) destabilize the population, consistently producing extinction with no mining adaptation.

F.5 NETWORK SIZE

We study how the size and architecture of the agent policy network affect mining and extinction frequency, and how environmental scale modulates this effect. We investigate 256×256 and 512×512 **Beach** worlds with initial populations of 2048 and 8192 (**RC**) agents, respectively (scaling four-fold with the number of grid cells), and run (**R**) baselines for each experiment, which again yield rare-to-nonexistent mining and 100% extinction rates.

The main experiments in Section 4 used agent policies with 2 layers and 64 channels, comprising 22465 parameters per agent (Table 11). Figure 19 shows that in the 256×256 **Beach**, holding the number of layers fixed while halving the channels to 32 (9185 total parameters) or doubling them to 128 (61313 parameters) has minimal-to-no effect on mining and extinction frequency. Network depth, in contrast, has a pronounced effect: every population of agents with just 1 layer and 32 channels went extinct, as did 50% of populations with 3 layers and 128 channels.

Crucially, scaling the environment to the 512×512 grid size produced markedly different dynamics for networks of varying depth (Figure 20). In the larger world, agent populations with 3-layer, 128-channel networks consistently mined and survived through 2M steps, and, strikingly, populations with the small 1-layer, 32-channel networks mined consistently as well. We hypothesize that both smaller and larger networks struggle to discover new behavior, but for distinct reasons: a smaller network’s reduced capacity makes good solutions hard to find, whereas a larger network’s expanded search space makes promising new behavior slower to locate. Regardless of the underlying cause, the fact that these networks fail so often in small environments yet succeed in larger ones further

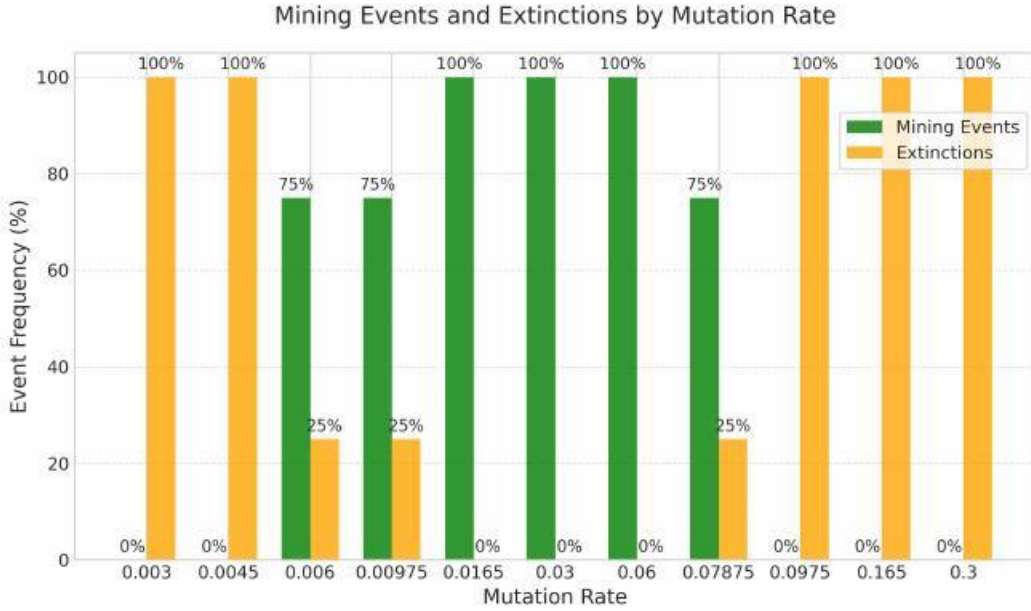


Figure 18: Frequency of mining and extinction events in a 256×256 **Beach** world across mutation rates. Frequencies are computed over four random seeds per setting.

illustrates the stability conferred by increasing scale. A larger environment sustains larger populations, affording greater exploration of policy space; at the same time, locally successful mutants propagate more slowly across a large environment, which promotes diversity and thereby again ensures greater exploration. For further discussion of these effects, see (Mitchell et al., 2006).

Table 11: Total parameter counts of (RC) and (R) agent policies across network architectures.

Architecture (layers, channels)	Parameter Count	
	(RC)	(R)
1, 32	8129	7969
2, 32	9185	9025
2, 64	22465	22145
2, 128	61313	60673
3, 128	77825	77185

F.6 VARYING VISION RANGE

We assess how vision range and viewing configuration affect predation and foraging behaviors in the 512×512 **Ocean** world with (RCV+A) agents. We evaluate visual sensor ranges of 3×3 , 7×7 , and 9×9 , and for each range compare the two viewing configurations shown in Figure 21 (a) Around, in which agents perceive grid cells in all directions, and (b) Front, in which agents cannot perceive cells behind them. The two configurations hold the total number of cells in the agent’s vision range equal. The main vision experiments in Section 4.2 used the 7×7 Around setting.

These results indicate that a small 3×3 vision range yields behavior largely indistinguishable from no vision at all (Blind), whereas the 9×9 range performs comparably to 7×7 —evidence of diminishing returns to larger vision ranges under the current environmental configuration. This is intuitive: because the agents are memoryless, visual features at the periphery of a 9×9 window are unlikely to alter the optimal action relative to a 7×7 window consistently enough to furnish a strong learning signal.

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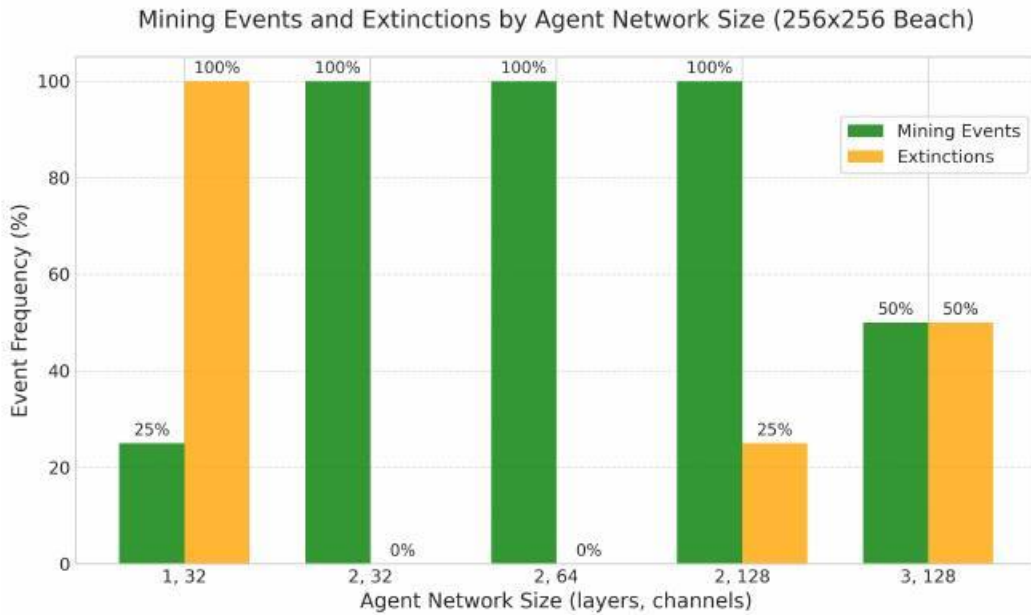


Figure 19: Frequency of mining and extinction events in a 256×256 **Beach** world across agent policy network sizes. Frequencies are computed over four random seeds per setting.

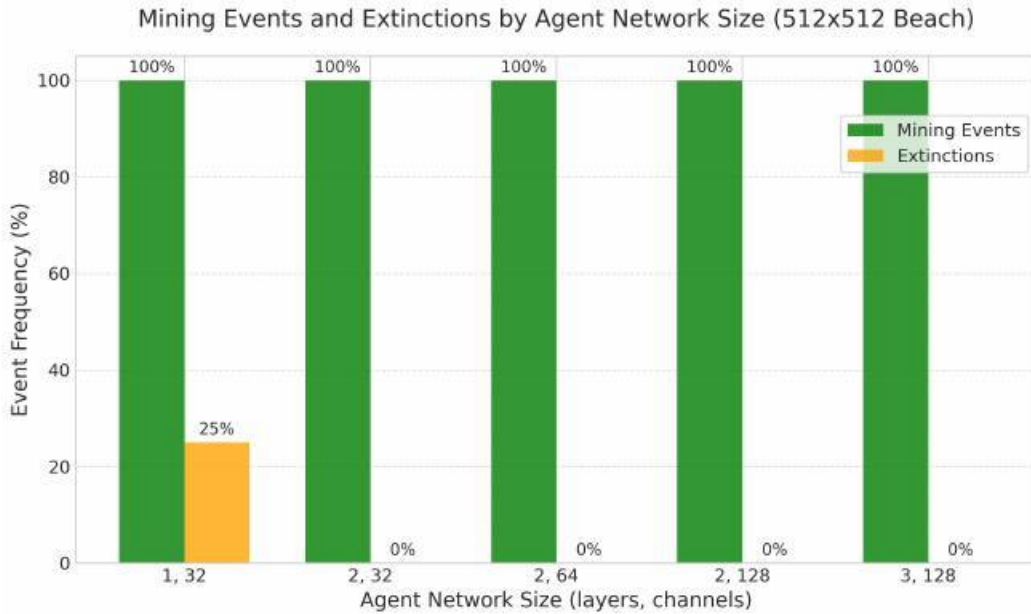


Figure 20: Frequency of mining and extinction events in a 512×512 **Beach** world across agent policy network sizes. Frequencies are computed over four random seeds per setting.

The data further suggest that at the 7×7 and 9×9 vision ranges, the Around configuration is slightly more favorable for population stability and for the effectiveness of attack and eat actions, delivering higher average population 🧑, biomass utilization ⚙️, and homicides per attack 🎯 than the Front configuration. We again attribute this to diminishing returns from perceiving more distant pixels: in the Around configuration, a greater share of the visible cells are nearby and thus likely to carry more information relevant to near-term action than the distant cells visible under the Front

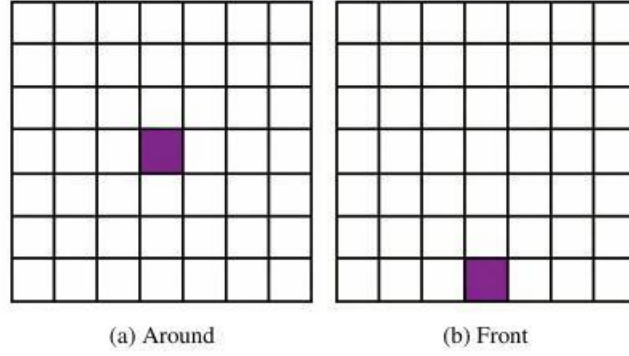


Figure 21: Comparison of the Around and Front viewing configurations for the 7×7 vision range. The purple cell denotes an agent, oriented toward the top of the page.

Table 12: Foraging and predation data in the 512×512 **Ocean** world for vision agents (**RCV+A**) across vision ranges and viewing configurations. See Tables 5 and 6 for the definitions of the metrics denoted by each icon. For each seed, every metric is averaged over all steps from step 500k onward, allowing the metrics to stabilize; reported values are the mean across the four seeds together with the standard deviation of the four seed averages.

Vision Range	Viewing Configuration						
3×3	Blind	7212 ± 2496	$.181 \pm .060$	$.435 \pm .090$	$.392 \pm .033$	$.032 \pm .006$	$.307 \pm .090$
	Around	10308 ± 4016	$.258 \pm .105$	$.352 \pm .014$	$.427 \pm .022$	$.040 \pm .030$	$.388 \pm .195$
	Front	13863 ± 520	$.351 \pm .023$	$.336 \pm .008$	$.437 \pm .009$	$.026 \pm .006$	$.470 \pm .145$
7×7	Around	16739 ± 122	$.373 \pm .003$	$.259 \pm .000$	$.532 \pm .001$	$.015 \pm .000$	$.751 \pm .007$
	Front	15255 ± 623	$.343 \pm .015$	$.253 \pm .001$	$.540 \pm .004$	$.015 \pm .000$	$.690 \pm .041$
9×9	Around	16744 ± 117	$.372 \pm .002$	$.257 \pm .001$	$.534 \pm .001$	$.015 \pm .000$	$.728 \pm .009$
	Front	15211 ± 1416	$.342 \pm .033$	$.253 \pm .001$	$.541 \pm .007$	$.015 \pm .000$	$.664 \pm .050$

configuration. At the 3×3 vision range, however, the trend reverses and the Front configuration becomes more favorable than Around, indicating that in this environment the marginal benefit of seeing one additional cell ahead under 3×3 Front outweighs that of seeing one cell behind under 3×3 Around, whose behavior most closely resembles the Blind setting.

F.7 VARYING ATTACK STRENGTH

We study how reducing agent attack strength affects emergent behavior and population dynamics in the 256×256 **Ocean** world. We evaluate three attack strengths: (1) Full Attack, dealing 10 HP of damage and instantly killing any agent within the attack radius; (2) Half Attack, dealing 5 HP; and (3) Quarter Attack, dealing 2.5 HP. The main **Ocean** experiments in Section 4.2 used the Full Attack setting.

Reducing attack strength from Full to Half markedly diminishes both the effectiveness and the frequency of attacks: attacks per agent and homicides per attack both drop considerably (Figure 22). As a consequence, populations under lower attack strength grow larger, with agents moving and eating more frequently and thereby attaining greater biomass utilization. Reducing strength further to the Quarter Attack setting yields diminishing returns, whether in the reduction of violence or the facilitation of additional population growth. The mechanism is a compounding one: when each attack deals less damage, a kill requires more attacks per agent, lowering the lethality of each attack; but as lethality falls, attacking becomes less useful and hence less likely to be selected for, so agents with lower attack strengths never become as skillful at attacking and instead devote more time to foraging.

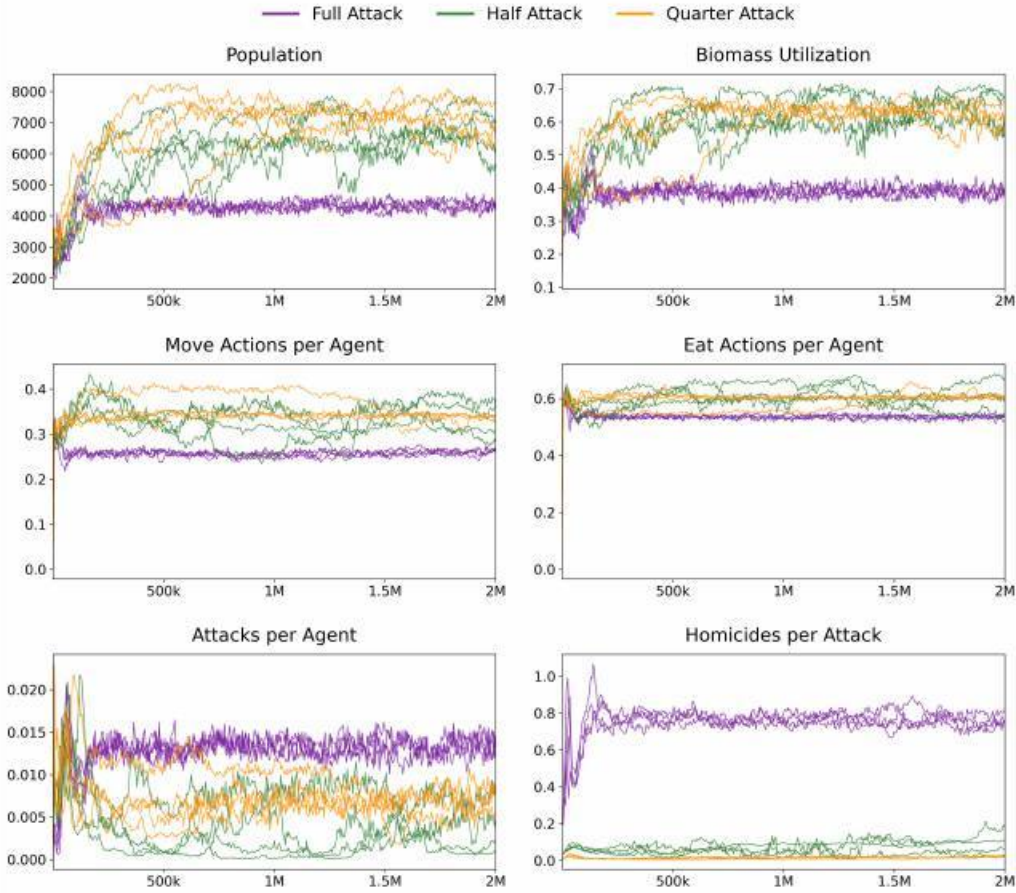


Figure 22: Population and action-frequency data over 2M steps in a 256×256 **Ocean** world for (RCV+A) agents across the Full Attack, Half Attack, and Quarter Attack strength settings. Four seeds are plotted per setting.

F.8 VARYING AGENT HP

We investigate the effect of reducing child agents’ starting HP and lowering the healing rate, which imposes pressure on agents to ‘grow’ over their lifetime in order to survive. The healing rate parameter determines how much HP an agent regenerates per healing step by converting available energy into HP; agents can therefore influence healing only by maintaining sufficient energy.

We study two settings: (1) Full HP, in which child HP starts at 5 and the healing rate is 1 HP per 0.1 energy; and (2) Low HP, in which child HP starts at 1 and the healing rate is 0.5. Maximum agent HP is 10 in all settings, and the main experiments (Section 4) used the Full HP setting.

Across the two settings, homicides per attack, eat-action frequency, and biomass utilization exhibit no significant difference (Figure 23). Under Full HP, populations grow slightly larger; under Low HP, agents attack less frequently and move slightly more. We hypothesize that agents born with lower HP must concentrate on foraging early in life in order to grow to full size, which would account for this difference.

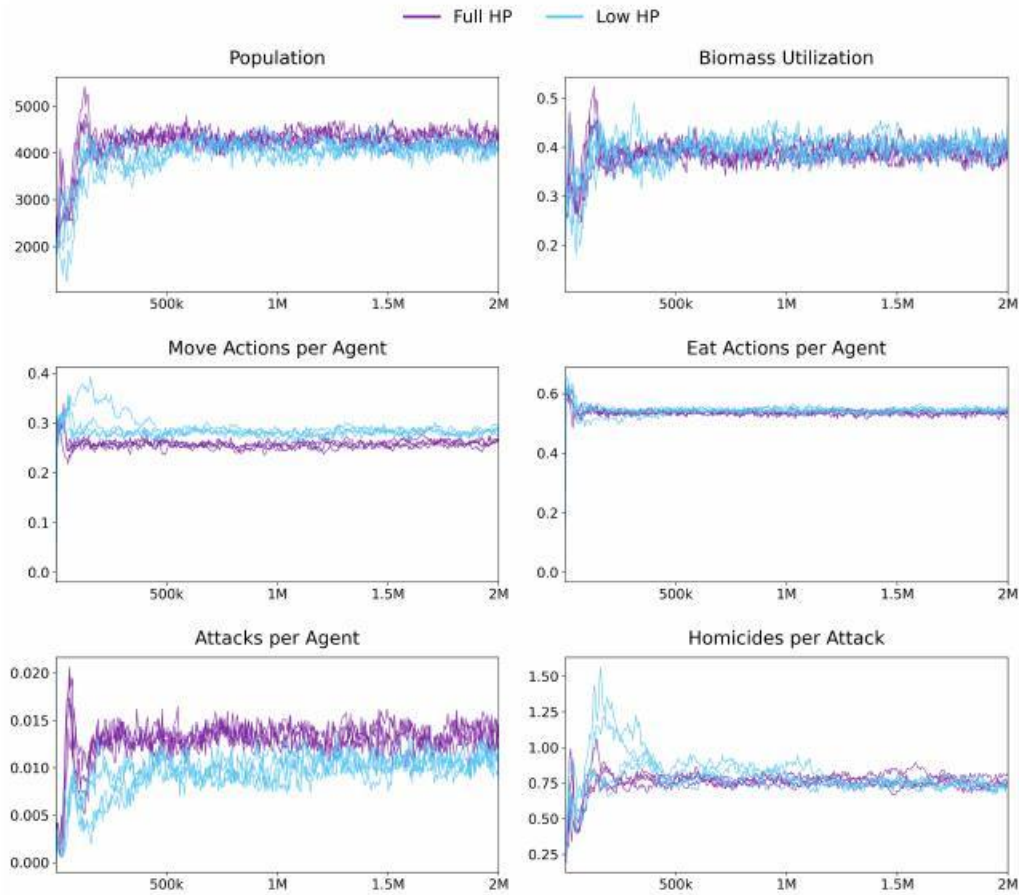


Figure 23: Population and action-frequency data over 2M steps in the 256×256 **Ocean** world for (RCV+A) agents across the Full HP and Low HP settings. Four seeds are plotted per setting.